

Charu Kanniga MP

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Summary

Computer Science Engineer passionate about Artificial Intelligence and Software Development, with hands-on experience building intelligent, scalable applications. Proficient in Python, SQL, and MongoDB, with practical knowledge of Machine Learning, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Chatbase, and AI-powered application development. Experienced in developing full-stack solutions, integrating AI APIs, and building data-driven systems using modern software engineering practices. Strong analytical, problem-solving, and collaborative abilities, committed to delivering innovative AI solutions that solve real-world challenges.

Education

Bachelor of Engineering in Computer Science Engineering — CGPA: 8.24 National Engineering College, Kovilpatti	2023 – 2027
HSE in Maths & Computer Science — 82% Our Own Modern Matric Higher Secondary School, Tirunelveli	2022 – 2023

Technical Skills

Programming Languages: C++, Python

AI & Machine Learning: Machine Learning, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Chatbase, Computer Vision, YOLO, OpenCV, Scikit-learn, Pandas, NumPy, Matplotlib

Databases: PostgreSQL, MongoDB

Tools & Platforms: Git, GitHub, Visual Studio Code, Jupyter Notebook, Streamlit, Docker, Jenkins

Core Concepts: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), DBMS

Soft Skills: Problem Solving, Analytical Thinking, Teamwork, Communication, Leadership, Time Management

Projects

DEBT DETOX — AI-Powered Financial Contract Auditor

Tech Stack: Python, Streamlit, Gemini API, Ollama/Gemma, Tesseract OCR, PyMuPDF, Pydantic, NumPy, SciPy, Plotly

- Built an AI-driven system that extracts and verifies financial terms from loan documents, detects hidden costs, calculates the effective borrowing cost, and provides evidence-backed repayment insights through OCR, deterministic financial computation, and document-grounded AI.

AI-Powered Company Recommendation System

Tech Stack: Python, Scikit-learn, Pandas, NumPy, Matplotlib, Flask, PostgreSQL, Git, GitHub

- Engineered an ML-based recommendation system that analyzes student performance, skills, certifications, and projects to surface optimal placement opportunities.
- Architected data preprocessing pipelines, feature engineering workflows, and predictive modeling to generate personalized recommendations and accelerate placement decision-making.

Real-Time Speech to Sign Language Translation

Tech Stack: Python, Machine Learning, SpeechRecognition, NumPy, Pandas, Flask, Git, GitHub

- Developed an assistive AI system that converts spoken language into sign language animations in real time, expanding communication accessibility for hearing-impaired users.
- Integrated speech recognition and ML techniques to generate accurate sign language output with minimal latency.

Internships

Thazhal Geospatial Analytics Pvt. Ltd. | Video-Based AI Intern | Aug 2025 – Oct 2025 (2 Months)

Tech Stack: Python, YOLO, OpenCV, NumPy, Computer Vision

- Developed AI-based video analytics solutions using YOLO and OpenCV for real-time object detection and tracking.
- Performed data preprocessing, model evaluation, and optimization to improve detection accuracy and system performance.
- Collaborated with the AI team to build scalable computer vision applications.

CodeSoft | Machine Learning Intern | Mar 2025 (1 Month)

Tech Stack: Python, Pandas, NumPy, Scikit-learn, Matplotlib

- Built and evaluated machine learning models for classification and prediction using Python and Scikit-learn.
- Performed data preprocessing, feature engineering, and visualization to improve model performance.

Achievements & Certifications

- Recipient of the **Best Paper Award** at the 2026 International Conference on Advanced Computing and Knowledge Engineering (ICACKE-2026), organized by IEEE Bangalore Section, IEEE Computer Society Bangalore Chapter, Institution's Innovation Council (IIC), and GITAM, for the research paper "AI-Driven High-Precision Mapping of Underground Pipelines using Multiset AI."
- Recognized as an **NPTEL Discipline Star** for outstanding academic performance and excellence across multiple technology courses.
- Achieved **Semi-Finalist** status in the ET-AI Hackathon 2026 conducted by The Economic Times, competing among top engineering teams in Artificial Intelligence and emerging technologies.
- Secured **First Prize** (INR 1,000 Cash Award) in Biztambola at Sethu Institute of Technology, demonstrating strong analytical thinking, logical reasoning, and problem-solving abilities.
- Earned **Snowflake Certification**, demonstrating proficiency in cloud data warehousing and modern data engineering concepts.
- Earned **MongoDB Certification**, validating expertise in NoSQL database design, data modeling, and database management.